



- 15 The height of each of 140 basketball players is recorded.
The table shows the results.

	$6 = 70$	$5 = 5$	$5 = 5$	$90 = 90$	$50 = 50$
Height (h cm)	$160 < h \leq 180$	$180 < h \leq 185$	$185 < h \leq 190$	$190 < h \leq 200$	$200 < h \leq 210$
Frequency	7	12	31	70	20

- (a) Calculate an estimate of the mean height.

M	170	182.5	187.5	195	205	
f	7	12	31	70	20	= 140
$h \times f$	1190	2190	5812.5	13650	4100	= 27082.5

$$\text{mean} = \frac{\text{Total } hf}{\text{Total } f} = \frac{27082.5}{140}$$

$$= 193.45$$

$$\dots\dots\dots 193.45 \dots\dots\dots \text{cm} [4]$$

- (b) Two of the players are chosen at random.

Find the probability that both players have a height greater than 190 cm and no more than 200 cm.

$$= \frac{70}{140} \times \frac{69}{139}$$

$$= \frac{69}{278}$$

$$\dots\dots\dots \frac{69}{278} \dots\dots\dots [2]$$

$$f_d = \frac{7}{20}$$

$$= 0.35$$

$$f_d = \frac{12}{5}$$

$$= 2.4$$

$$f_d = \frac{31}{5}$$

$$= 6.2$$

$$f_d = \frac{70}{20}$$

$$= 3.5$$

$$f_d = \frac{20}{20}$$

$$= 1$$

DO NOT WRITE IN THIS MARGIN

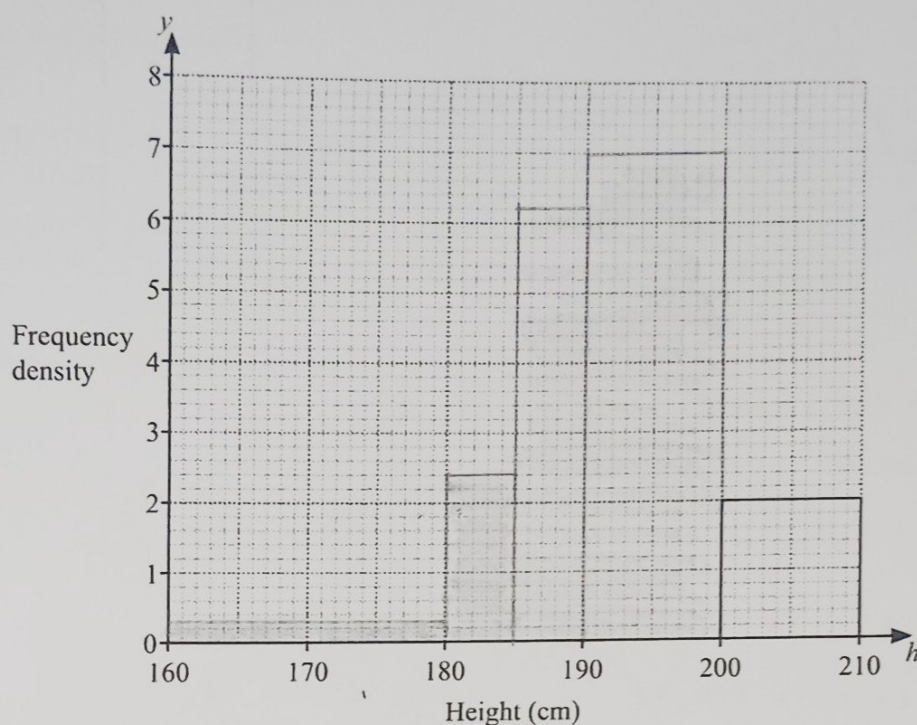
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(c) Complete the histogram to show the information in the frequency table.



[3]

- 16 Mateo invests \$1250 at a rate of $r\%$ per year compound interest. At the end of 6 years the total value of his investment is \$1484.

Calculate the value of r .

$$2.9\% = r$$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$\frac{1484}{1250} = \frac{1250}{1250} \left(1 + \frac{r}{100} \right)^6$$

$$1.1872 = \left(1 + \frac{r}{100} \right)^6$$

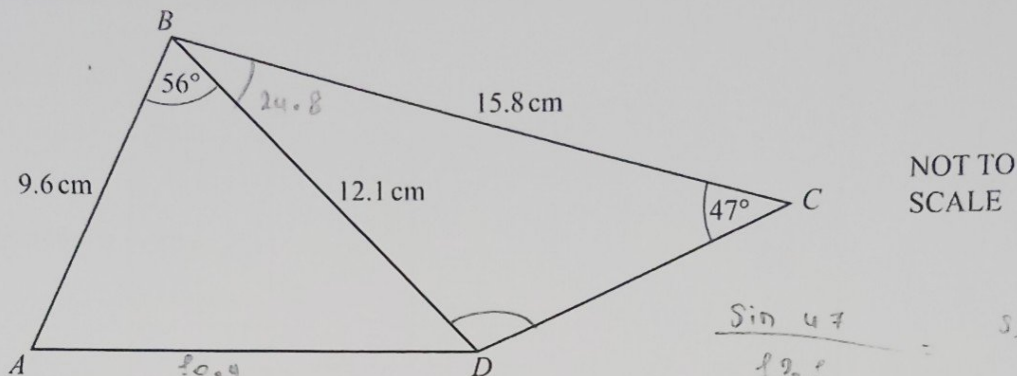
$$\sqrt[6]{1.1872} = 1 + \frac{r}{100}$$

$$r = \dots\dots\dots 2.9\% \dots\dots\dots [3]$$

$$\left(\sqrt[6]{1.1872} \right) - 1 = \frac{r}{100}$$

$$\left[\left(\sqrt[6]{1.1872} \right) - 1 \right] \times 100 = r$$

17



The diagram shows a quadrilateral $ABCD$.

(a) Calculate AD .

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 9.6^2 + 24.8^2 - 2 \times 9.6 \times 24.8 \cos 56$$

$$c^2 = 238.57 - 109.91$$

$$c^2 = 108.66$$

$$c = \sqrt{108.66}$$

$$c = 10.4 \text{ cm}$$

$$AD = 10.4 \text{ cm [3]}$$

(b) Calculate the obtuse angle BDC .

$$\frac{\sin 47}{12.1} = \frac{\sin \theta}{25.8}$$

$$\sin^{-1}(0.95) = \theta \quad \text{working on page}$$

$$79.8 = \theta$$

$$\frac{\sin 47}{12.1} \times 15.8 = \sin \theta$$

$$180 - 79.8 = 100.2$$

$$0.95 = \sin \theta$$

$$\text{Angle } BDC = 100.2^\circ \text{ [4]}$$

(c) Calculate the area of the quadrilateral.

$$= \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 24.8 \times 9.6 \sin 56$$

$$= 48.2 \text{ cm}^2$$

$$= \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 15.8 \times 12.1 \sin 24.8$$

$$= 41.4 \text{ cm}^2$$

$$89.6 \text{ cm}^2 \text{ [3]}$$

$$= 41.4 + 48.2$$

$$= 89.6 \text{ cm}^2$$

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN